

Rishi Das

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EDUCATION/ACADEMIC BACKGROUND

NIT Rourkela

B.Tech.+M.Tech.(Dual Degree), Mining [GPA: 8.39/10.0]

Rourkela, Odisha

Nov. 2022 – June 2027

Salt Lake School(E.M)

10+2 High School Education [Grade: 10.0]

Kolkata, West Bengal

April 2008 – June 2022

RESEARCH EXPERIENCE

Undergraduate Research Intern

IIT Kharagpur

May 2025 – August 2025

Kharagpur, West Bengal

- On-site internship entitled '**Embedded Systems Design and Coding**' under [Professor Arabinda Routray](#) at IIT KGP during the mentioned period. I worked on a research project titled "**Contactless Monitoring of Human Vitals and Emotions**" based on fluctuations in heart rate, respiratory rate, and other vital parameters, determined through non-invasive biomarkers using FMCW radar and thermal camera images. [Project Certificate](#)

Undergraduate Research Assistant

NIT Rourkela

Nov.2024 – Present

Rourkela, Odisha

- Designed and implemented a hybrid architecture combining convolutional and transformer-based models with feature fusion and optimisation techniques. My advisor and guide for the project is [Professor Saptarshi Chatterjee](#).
- Developed a custom CVTLeukemia model for early ALL classification using the **ISBI C-NMC 2019 dataset (15k images)**. Built a preprocessing pipeline with colour-space thresholding, morphological filtering, and nucleus segmentation to enhance domain generalisation. Implemented **Sparse Self-Attention, Squeeze-and-Excitation blocks, hierarchical token embeddings, and DropBlock regularisation**, achieving 80 per cent accuracy with strong precision-recall performance. Trained using AdamW + OneCycleLR with extensive augmentation (colour jitter, rotations, normalisation). Designed a full hardware deployment using a Raspberry Pi 4B, microscope-mounted Pi Camera, and a Dockerised Flask-based REST API for real-time screening—enabling low-cost, scalable CAD diagnostics for rural healthcare settings.

NeurIPS 2024 Conference Participant

Remote

Apr.2024 - Jul.2024

Leash BELKA Dataset- Kaggle Challenge

- Participated in the **NeurIPS 2024 "Predict New Medicines with BELKA" competition** to develop machine learning models for predicting small molecule-protein binding affinity, a critical step in accelerating drug discovery. Tackled the challenge using DNA-encoded chemical library (DEL) data, represented by SMILES strings, to classify binding behaviour across three protein targets. Explored and implemented various methodologies, including **ChemBERTa, Boosting models, single-dimension CNNs, decision trees, and ensemble techniques**.
- Achieved a **top 5 per cent rank on the public leaderboard and a score of 0.26367 in private test data**, demonstrating the efficacy of our solutions. We ranked among the **top 12 per cent of the overall participants**. [Leaderboard](#)

ISIC 2024 - Skin Cancer Detection with 3D-TBP

Remote

Feb. 2024 - Apr. 2024

Independent Research

- Contributed to the **ISIC 2024 challenge on early skin cancer detection using image-based algorithms for 3D total body photographs**. The goal was to accurately classify benign and malignant lesions, aiding triage in underserved populations. Engaged in extensive data interpretation, preprocessing, augmentation, and feature extraction from diagnostically labelled images and associated metadata. Researched public codes and models to develop a unique classification model incorporating linear neural nets, global pooling and sigmoid activation.
- Evaluated model performance using **Binary AUROC**, for precise diagnostic predictions. This work deepened my knowledge of AUROC metrics, understanding machine learning concepts specific to medical image analysis. and their implementation for real-world medical datasets. This is the [ISIC Code](#) that I implemented and got [this RESULT](#).

PROJECTS

Anime Face Generator | *Python, PyTorch, GANs*

Sept. 2024 – Oct. 2024

- Implemented a simplified Generative Adversarial Network (GAN) to generate anime faces using the public [AnimeFaceDataset](#) in a [Kaggle Notebook](#) powered by a P100 GPU.
- Trained the model over 20 epochs, achieving an effective accuracy score and generating visually appealing results. Published a [Medium post](#) explaining GAN concepts in an accessible manner, which received positive feedback for its clarity. The notebook earned a **Kaggle bronze medal with 500+ views** and widespread appreciation for its beginner-friendly approach.

Bhashamitra AI | *HTML, TailwindCSS, ReactJS, HuggingFace APIs, Flask, PyTorch, Python*

April 2025

- Deployed real-time audio→text translation using Whisper API with Flask backend and React frontend; containerised with Docker. Secured 1st Runners-up after two evaluation rounds by Reverie Language Technologies. [Project Repository](#)

Cervello | *HTML, TailwindCSS, FastAPI, OpenAI API, Qdrant, MongoDB, PyTorch, Python*

March 2025

- Developed a real-time RAG application using OpenAI API and Qdrant vector DB for multimodal enterprise document retrieval, query handling, and support ticketing on software-based issues. Secured 2nd position at HackFest 2025 at IIT(ISM) Dhanbad with **TEAM NITROUS**. [Project Repository](#)

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, HTML/CSS, MongoDB, SQLite, PostgreSQL

Frameworks/Libraries: Flask, Hibernate, JDBC, SpringBoot, ReactJS, LangChain, Bcrypt, Pandas, NumPy, Matplotlib, SciPy, Seaborn, scikit-learn, PyTorch, QGIS, MATLAB

Developer Tools: GitHub, Git, Jupyter Notebook, GCP, Docker, Maven, Postman, VSCode, Eclipse

AWARDS/HONORS/ACHIEVEMENTS

- StatusCode1.0: Bagged the **2nd runner-up prize** in open track in a 30-hour-long hackathon at IIIT Kalyani, West Bengal, India.
- Kaggle Contributor** : Several Badges and medals were achieved through participation in the competition, codes, and discussion work on data science.
- 50 Days Badge Leetcode 2024** : Earned the 50 Days Batch on Leetcode, widely considered the top public platform for Data Structure and Algorithm practice. This badge is awarded to the top 7 per cent of the consistent coders in the platform.
- Udemy Certificate** : Completed a 15+ hours Udemy course on complete Python BootCamp, including basics, loops, data structures, web scraping, libraries, tkinter, etc.
- NITRIMUN 2023** : Bagged the best delegate award at NIT Rourkela International Model United Nations 2023 by representing India at the ECOSOC Committee.

LEADERSHIP / EXTRACURRICULAR

Vice President

May 2025 – Present

Machine Learning for Everyone

NIT Rourkela

- Organise hackathons, build AI/ML projects, drive research and open-source, and conduct sessions on AI development.

Secretary

May 2024 – April 2025

Finance Club

NIT Rourkela

- Executed 5 large events (e.g., Finotsav); secured major sponsors and collaborations.